

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2021

ESE 4003

(Hydrogen and Fuel Cells)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

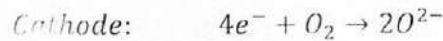
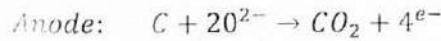
SECTION – A

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|-------|---|----|
| 1(a). | What are the sources of hydrogen fuel? Explain environmental implications of hydrogen fuel. | 08 |
| 1(b). | Explain Honda-Fujishima effect. How Honda-Fujishima effect can be used for hydrogen production? Explain with proper sketch. | 15 |
| 1(c). | Delineate photocatalysis. What are the factors affecting photocatalytic activity? | 12 |
| 2(a). | What is meant by biomass gasification? Explain hydrogen production by biomass gasification. | 12 |
| 2(b). | What are the main hydrogen evolving enzymes? Explain how these enzymes work for biohydrogen production. | 14 |
| 2(c). | Explain the working principle of pressure swing adsorption for hydrogen purification. | 09 |
| 3(a). | Why is it necessary to redesign the fuel delivery system of an internal combustion engine for hydrogen fuel? Briefly discuss any one method of hydrogen fuel delivery system. | 12 |
| 3(b). | Discuss the operations and maintenance of hydrogen transmission through pipeline. | 11 |
| 3(c). | What are the advantages of blending hydrogen with natural gas for transmission? What are the challenges associated with the blending of hydrogen with natural gas? | 12 |
| 4(a). | Discuss the benefits and barriers of different hydrogen storage technologies. | 10 |
| 4(b). | What are the challenges of using hydrogen as a fuel for internal combustion engine? Why crankcase ventilation is necessary for hydrogen internal combustion engine? | 12 |
| 4(c). | Interpret the properties of gaseous and liquid hydrogen responsible for safety issues. | 13 |

SECTION – B

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|-------|---|----|
| 5(a). | Being an energy engineer, why you should study the fuel cell technology? | 08 |
| 5(b). | What are the limitations of Polymer Electrolyte Membrane Fuel Cell? Discuss the characteristics of fuel cell charge transport resistance. | 15 |
| 5(c). | Write down the differences between battery and fuel cell. Describe the fabrication methods for gas diffusion electrodes. | 12 |

- 6(a). Explain the integrated gasification carbonate fuel cell concept. 10
- 6(b). What is MEA? Enumerate the anodic, cathodic, and overall cell reactions for DMFC. 10
- 6(c). A fuel cell has the following reactions: 15



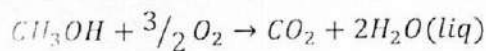
At RTP, the changes of enthalpy and free energy, per kilo-mole of CO_2 are: $\Delta h_f^0 = -393.5 \text{ MJ}$ and $\Delta g_f^0 = -394.5 \text{ MJ}$. What is the overall reaction? What is the ideal emf (i.e., V_{rev})? What is the difference in entropy between reactants and products (assuming that the cell operates isothermally)?

- 7(a). Describe fuel cell thermodynamic efficiency. Show that for a H_2/O_2 fuel cell operating under thermodynamically reversible conditions, 15

$$\left(\frac{\partial E}{\partial P}\right)_T = -\frac{\Delta n_g RT}{nF_p}$$

- 7(b). Discuss the emissions from fuel cell power plants. What are the environmental benefits and drawbacks associated with fuel cell power plant? 10
- 7(c). A fuel cell operates for 1 hour at 2 A current load and then operates for 2 more hours at 5 A current load. Calculate the total number of moles of H_2 consumed by the fuel cell over the course of this operation. To what mass of H_2 does this correspond? Given that molar mass of hydrogen = 2 g/mol. Assume 100% fuel utilization. 10

- 8(a). A direct methanol fuel cell uses methanol as fuel. Where, temperature is 30°C and pressure is 100 kPa. Calculate: (i) Δh_{rxn}^0 , (ii) Δs_{rxn}^0 and (iii) Δg_{rxn}^0 . 09



Chemical Species	Δh_f^0 (kJ/mol)	Δs_f^0 (J/mol.K)
CH_3OH	-200.95	239.83
O_2	0	205.14
CO_2	-393.51	213.8
H_2O (liq)	-285.83	69.95

- 8(b). What is cogeneration? Explicate the design of a combined heat and power (CHP) fuel cell system with necessary diagram. 14
- 8(c). How can the kinetic performance of a fuel cell be improved? Discuss. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2021

ESE 4213

(Nuclear Power Engineering)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Write short note on the following terms: 06
(i) Fission chain reaction
(ii) Fissile material
(iii) Fertile material
- 1(b). “As the mass number increases, stable nuclides have an increasing ratio of neutrons to protons”. Justify the statement with necessary figure. 09
- 1(c). Find the mass defect, total binding energy and binding energy per nucleon of ${}^{62}_{28}\text{Ni}$. The neutral atomic mass of ${}^{62}_{28}\text{Ni}$ is $61.928349u$. 07
- 1(d). What is meant by nuclear binding energy? Briefly describe it with necessary diagram. 13
- 2(a). Discuss about the environmental aspects of nuclear power generation. 12
- 2(b). What are the fundamentals safety objectives according to the IAEA? Also, what are the fundamentals safety requirements? 11
- 2(c). Describe about the importance of leadership and management to attain safety of a nuclear power plant. 12
- 3(a). How Uranium can be extracted from the primary ore? Briefly describe the ISL mining process. 12
- 3(b). What is fuel enrichment? With the help of schematic diagram, describe the gas centrifuge method of fuel enrichment. 11
- 3(c). How fuel assembly is made from fuel pellet? Describe with neat sketch. 12
- 4(a). With the help of necessary diagram, describe the reprocessing process of used nuclear fuel. 15
- 4(b). What are the several ways to dispose radioactive waste? Describe near-surface disposal procedure with necessary figures. 10
- 4(c). List the interim storage facilities of radioactive waste. Describe the storage pond. 10

SECTION – B

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|-------|--|----|
| 5(a). | With the necessary diagram, explain the working principle of a nuclear power plant. How it is different from other conventional thermal power plant? | 12 |
| 5(b). | What are the different types of nuclear reactor? Which type of reactor is mostly used all around the world and why? | 08 |
| 5(c). | Can nuclear power become the future of energy source to meet the world energy demand? Explain in context of global. | 09 |
| 5(d). | Write down the functions of coolant and pressure vessel in nuclear power plant. | 06 |
| | | |
| 6(a). | What are the specific features of Generation III+ reactors? How it differs from Generation IV technologies? | 10 |
| 6(b). | What is meant by EPR? Describe the special features of EPR design. | 12 |
| 6(c). | With the help of schematic diagram, describe the layout of a PWR plant. | 13 |
| | | |
| 7(a). | What is meant by NSSS? Describe it briefly. | 12 |
| 7(b). | With neat sketch, describe the working principle of Supercritical-Water-Cooled Reactor. | 12 |
| 7(c). | What are the major components and functions of lower core support structure of a nuclear reactor? Also write down the differences between core barrel and core baffle. | 11 |
| | | |
| 8(a). | Define fuel pallets. Describe the process of fuel rod formation. | 12 |
| 8(b). | What are the different patterns to place fuel assemblies in reactor core? Describe. | 12 |
| 8(c). | What is RCCA? Briefly explain its function and formation. | 11 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2021

ESE 4017

(Energy Efficiency Assessment)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by energy performance assessment? What are the different factors that affect boiler performance? 08
- 1(b). What are the different ways to evaluate boiler efficiency? What are the different parameters to be considered in indirect method? 07
- 1(c). The following are the data collected for a boiler using furnace oil as the fuel. Find out the boiler efficiency by indirect method. 20
- Fuel firing rate = 2500 kg/hr
- Surface temperature of boiler = 90°C
- Surface area of boiler = 100 m²
- Humidity = 0.03 kg/kg of dry air
- Wind speed = 4 m/s
- Fuel analysis (%):
- Carbon = 84
- Hydrogen = 12
- Nitrogen = 0.5
- Oxygen = 1.5
- Sulphur = 1.5
- Moisture = 0.5
- GCV of fuel = 1500 kCal/kg
- Flue gas analysis:
- Flue gas temperature = 210°C
- Ambient temperature = 30°C
- CO₂ percentage in flue gas by volume = 10.5
- O₂ percentage in flue gas by volume = 7.5
- 2(a). What is meant by blow down? Briefly describe about continuous blow down with necessary sketches. 10
- 2(b). Why the feed water of boiler needs to be treated? Discuss about the different impurities that contain in raw water. 12
- 2(c). Explain the coagulation-flocculation process of boiler feed water. 10
- 2(d). What is meant by deaeration? 03
- 3(a). Write down the differences between reverse osmosis and normal osmosis. What contaminants will reverse osmosis remove from water? 10
- 3(b). What are the different approaches to save the energy used in boiler? Explain how energy can be saved in boiler by followings: 10
- (i) Excess air control
- (ii) Proper boiler scheduling

- 3(c). What are the different sources of waste heat in boilers that can be recovered. Discuss the necessary information/steps needed to recovery of heat in boiler plant. 10
- 3(d). Discuss the direct and indirect benefits of waste heat recovery in boiler. 05
- 4(a). Write down the differences between system characteristics curve and fan curve. With necessary figure, explain how to determine the operating point of a fan. 08
- 4(b). Briefly explain the factors that should be considered to select a fan. 10
- 4(c). How energy can be saved by using a variable speed drive to control the air flow of a fan/blower? 05
- 4(d). A centrifugal fan has efficiency of 55%. From following figure, find the nearest approximation of (i) static pressure in mmWC, (ii) flow rate in m³/hr, (iii) shaft power in kW, (iv) work out, (v) power drawn by motor if the motor operating with an efficiency of 90%. 12

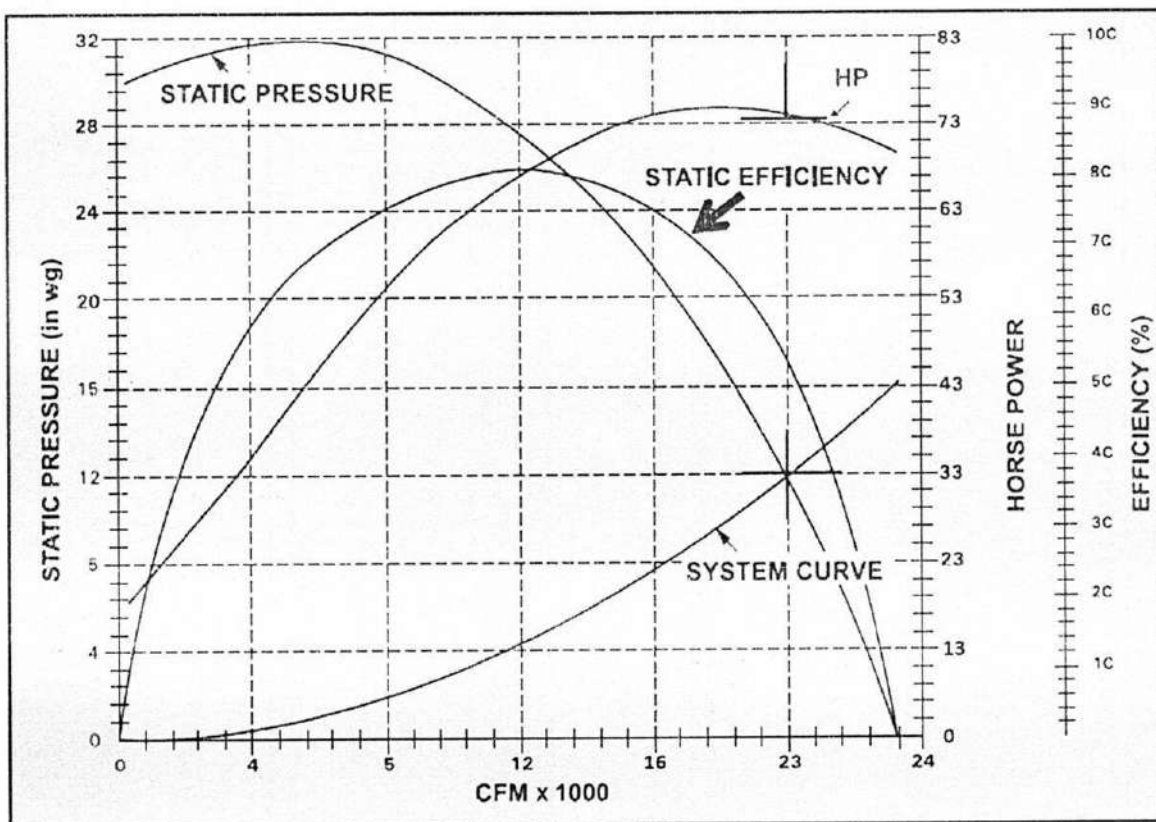


Figure: Fan Characteristic Curve

SECTION – B

- 5(a). Discuss the factors that affect the performance of pump. 10
- 5(b). Explain the various methods of pump capacity control. 10
- 5(c). A centrifugal clear water pump rated for 800 m³/hr was formed to be operating at 600 m³/hr with discharge valve throttled. The pump speed is 1500 RPM. The discharge pressure of the pump before the throttle valve is 3 kg/cm²g. The pump draws the water from a sump 5 meters below the centerline of the pump. The input power drawn by the motor is 120 kW at a motor efficiency of 88%. 15
- Find out the efficiency of the pump.
 - If the normal required water flow rate is 450 m³/hr to 600 m³/hr, what is your opinion should be the most energy efficient option to get the required flow rate variation?
 - What would be the pump shaft power for that most energy efficient option if the pump is delivering the flow rate of 525 m³/hr?

- 6(a). Write down the purposes of performance assessment of refrigeration plant. What are the basic parameters required to measure while estimating the chiller performance? 10
- 6(b). How the net refrigeration capacity at the evaporator and the heat rejection at the condenser of VCR system can be measured? Describe in brief. 10
- 6(c). How the performance of an air conditioning system can be evaluated? Explain. 07
- 6(d). An air handling unit was evaluated for its performance and the following parameters were recorded: 08
- The air velocity was measured at different locations of the face area of 1.2 m^2 for AHU unit and its average velocity was calculated as 1.5 m/s . The DBT and WBT of the air at the inlet of the unit are 29.5°C and 25°C , and at the outlet 25°C and 20°C , respectively. Using psychrometric chart, calculate the heat load in TR.
- 7(a). What are the different technical aspects of energy efficient motors? What is meant by motor characteristics? 10
- 7(b). Discuss about the approaches to improve efficiency of a motor. 08
- 7(c). What are the different technical losses in motors? What are the measures to reduce the technical losses? 07
- 7(d). Briefly explain the followings: 10
- (i) Voltage unbalance of motor
 - (ii) Over motoring
- 8(a). Why location of capacitors is important? Write down the advantages of proper installation of capacitors. 08
- 8(b). What are the sources of waste heat recovery in diesel generator system? Waste heat recovery from which source is more convenient and why? 10
- 8(c). Explain the significance of the followings on cooling tower performance: 12
- (i) Flow rate control
 - (ii) Cooling tower fans
 - (iii) Cooling tower operation
 - (iv) Effectiveness
- 8(d). Why VFD is more energy efficient method than control valves when it comes to control the flow rate of a pump? 05

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY**Department of Energy Science and Engineering**B. Sc. Engineering 4th Year 2nd Term Examination, 2021

ESE 4219

(Energy Audit and Management)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is energy audit? With example, describe preliminary energy audit. 09
- 1(b). List the various energy audit instruments. What are the functions of fyrite and ultrasonic steam trap tester? 08
- 1(c). How to identify energy conservation opportunities in energy audit? 06
- 1(d). What is meant by plant energy performance? An Energy Manager in a factory has collected following data: 12

Reference year (2021) energy use = 27 million kcal; production factor (PF) for the current year (2022) = 0.92; current year's energy use = 24 million kcal. Calculate the plant energy performance (PEP) of the factory for the year 2022. State your inference.

- 2(a). What is meant by energy benchmarking? How is it helpful? 07
- 2(b). What is payback period? What are the advantages and limitations of this method? 09
- 2(c). What is energy performance contracting? Explain its various types. 09
- 2(d). Distinguish between NPV and IRR. Calculate the internal rate of return for the following cash flow of a period. 10

Year	0	1	2	3	4
Cash flow	500,000	80,000	80,000	90,000	95,000

- 3(a). What is a project? What criteria should be considered while screening a project? 07
- 3(b). What is Gantt chart? What are its limitations? How it can be enhanced? 08
- 3(c). Why do manage of reactive power needed? How reactive power can be controlled? 05
- 3(d). Distinguish between CPM and PERT. WBS of a project is given below: 15

Activity	Immediate predecessor	Completion time (weeks)
A	-	7
B	A	3
C	B	1
D	A	8
E	D, C	2
F	D, C	1
G	D, C	1
H	F	3
I	H	2
J	E, G I	1

- (i) Draw the network diagram.
- (ii) Find out the critical path.

- 4(a). What factors affecting waste heat recovery? What are the various techniques to recover waste heat? 09
- 4(b). What modifications is proposed in BNBC for building energy efficiency? Describe the objectives of building energy and environment rating of SREDA. 08
- 4(c). What is energy performance index of building? Explain the main characteristics of green building. 09
- 4(d). Explain the energy conservation opportunities in boiler. 09

SECTION – B

- 5(a). Why material and energy balance is required? Explain the material and energy balance in unit operations. 08
- 5(b). How can energy be lost in a system? In a continuous centrifuging of milk, if 35,000 kg of whole milk containing 4% fat is to be separated in a 6 hour period into skim milk with 0.45% fat and cream with 45% fat, what is the flow rate of the two output streams from the continuous centrifuge which accomplishes the separation? 12
- 5(c). Explain the Sankey diagram. 05
- 5(d). What are the characteristics that a company must maintain while setting targets? With proper example, explain the differences between target and goal. 10
- 6(a). What is the most effective way to motivate employees for energy management program? Explain. 06
- 6(b). What is “KAIZEN”? What are the steps need to follow to implement “KAIZEN”? 10
- 6(c). What are the major areas that management needs to address during the training of the employees? Write down the name of six big losses that lower the equipment efficiency. 07
- 6(d). Steam pipe line 100 mm diameter is not insulated for 100 m length supplying steam at 10 kg/cm² to the equipment. Find out the fuel savings if it is properly insulated with 65 mm insulating material. Consider boiler efficiency 80%, fuel cost BDT 15000/ton, surface temperature without insulation 170°C, surface temperature after insulation 65°C, ambient temperature 25°C. 12
- 7(a). What is meant by TQM? What are the benefits of ISO 5001? 10
- 7(b). What is heat pipe? With a schematic diagram, describe the working principle of a heat pipe. 06
- 7(c). In a crude distillation unit of a refinery, furnace is operated to heat 500 m³/hr of crude oil from 255°C to 360°C by firing 3.4 tons/hr of fuel having GCV of 9850 kcal/kg. As an energy conservation measure, the management has installed an air preheater (APH) to reduce the flue gas heat loss. The APH is designed to preheat 57 tons/hr of combustion air to 195°C. Calculate the efficiency of the furnace before and after the installation of APH. Consider the following data:
Specific heat of crude oil = 0.6 kcal/kg°C, specific heat of air = 0.24 kcal/kg°C, specific gravity of crude oil = 0.85, ambient temperature = 28°C. 10
- 7(d). What is the short term and long term objective of UNFCCC? What is the major difference between protocol and convention? 09
- 8(a). What are the Kyoto mechanisms? How do these mechanisms help green investment and meet emission targets? 13
- 8(b). Write down the name of five potential CDM projects. 05
- 8(c). What is monitoring and targeting? What are the benefits of monitoring and targeting? 07
- 8(d). In a food processing plant, the monthly production related energy consumption was 1.8 times the production and non-production related energy consumption was 15000 kWh per month up to May 2010. In the month of June 2010, a series of energy conservation measures were implemented. Use CUMSUM technique to develop a table and calculate energy savings for subsequent 6 months period from the data given below: 10

Month	Production (kg)	Actual Energy Consumption (kWh)
July-2010	62000	113600
August-2010	71000	139000
September-2010	75000	158000
October-2010	59000	119300
November-2010	62000	123700
December-2010	73000	143600

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2021

ESE 4013

(Energy in Built Environment)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | How should building energy performance contribute to reduce the energy consumption? | 08 |
| 1(b). | Explain the energy use in building. What are the activities influence the energy use in buildings? | 09 |
| 1(c). | What is indoor environment control (IEQ) of a building? What are the internal and external factors upon which energy consumption of a building depend? What are the adverse effects of IEQ? | 11 |
| 1(d). | When developing a new building, what are the factors should be considered to reduce artificial energy consumption? | 07 |
| | | |
| 2(a). | What is visual comfort? Explain the design consideration for visual comfort of a building. | 08 |
| 2(b). | What are the major sources of indoor air contaminations? Describe the personal responsibility to reduce the indoor air contaminations. | 12 |
| 2(c). | What is meant by acoustic comfort? Explain the parameters that should be maintained for good acoustic condition in building. | 08 |
| 2(d). | Explain the energy used pattern in a building. | 07 |
| | | |
| 3(a). | Distinguish between weather and climate. Explain the effects of building orientation on the solar heat gain of a building. | 08 |
| 3(b). | What does mean by sun-earth relationship? With the help of necessary sketch, explain the significance of earth relation and revolution on building heat gain. | 11 |
| 3(c). | What is the relationship between ambient temperature and sky temperature? How sky temperature can be calculated? | 06 |
| 3(d). | Draw five wind flow patterns around a building which will increase the natural wind flow and mention the amount of increment. | 10 |
| | | |
| 4(a). | What are the different options of natural lighting in building? Among top and side daylighting, which one is preferable? Explain. | 08 |
| 4(b). | What is daylighting? What are the benefits and drawbacks of daylighting? | 06 |
| 4(c). | What are the advantages and disadvantages of sun shading devices? Explain the horizontal and vertical shadow angle to determine the dimension of shading device. | 12 |
| 4(d). | Write short notes on: (i) Saw tooth lighting, (ii) Tubular daylighting, (iii) Daylight reduction. | 09 |

SECTION – B

- 5(a). Explain the ASHRAE method of calculating heat losses from walls and floor of basement. Write the equation to calculate transient temperature of soil. 10
- 5(b). What is sol-air temperature? Why sol-air temperature differs from ambient temperature? 07
- 5(c). Why zoning of building is required? 05
- 5(d). The south masonry wall of a house is made of 10-cm-thick red face brick, 10-cm-thick common brick, 19-mm-thick air space, and 13-mm-thick gypsum board, and its overall heat transfer coefficient is $1.8 \text{ W/m}^2\cdot^\circ\text{C}$, which include the effect of convection on both sides. The cooling system is to be sized on the basis of the heat gain at 3 PM solar time and sol-air temperature in the location at that time is 48°C . The interior of the house is to be maintained at 22°C and the area of the wall is 20 m^2 . If the design ambient air temperature at that time is 35°C , determine:
- (i) the design heat gain through the wall,
 - (ii) the fraction of this heat gain due to solar heating.
- 6(a). Explain the typical characteristics of window glazing in cold and hot location with schematic diagram. 09
- 6(b). How does color affect the heat transfer through building walls? Explain. Explain the ways to minimize the air space between the glasses of double-pane window. 09
- 6(c). What is smart window? Describe the various types of smart windows. 09
- 6(d). Why building thermal load should be determined on transient condition? Explain the CTF model. 08
- 7(a). What are the major components of energy auditing of a building? Discuss about the economic goal of energy management in building? 11
- 7(b). How to measure the air infiltration rate in a building? Distinguish between crack method and air-change method for prediction of air infiltration. 08
- 7(c). What are the elements should be considered while estimate the annual energy consumption of a building in design stage. Explain the Bin method. 08
- 7(d). What are meant by heating degree-day and cooling degree-day? How to determine the maximum threshold temperature to switching between air-conditioning and ventilation only? 08
- 8(a). What is meant by nearly zero-energy building? Distinguish between net-zero site energy and net-zero source energy building. 09
- 8(b). Discuss the principles for designing a net-zero energy building. 09
- 8(c). Discuss about the challenges of technologies for low energy building. 08
- 8(d). Describe the evaporative cooling system with desiccant dehumidification. 09

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